



PIMS PRO

Pipeline Integrity Management System

Pipeline Corrosion Assessment & Remaining Life Calculator

Version	1.0
Developer	M. Fahad Zafar
Report File	PL-005_Inspection_Report_2026-07-16.pdf
Generation Date	16 July 2026

Inspection Information

Pipe ID	PL-005
Pipeline Name	PL-005
Material	Carbon Steel
Fluid	Crude Oil
Inspector	Ali
Inspection Date	2026-07-16
Operating Pressure	4.0 MPa

Engineering Calculations

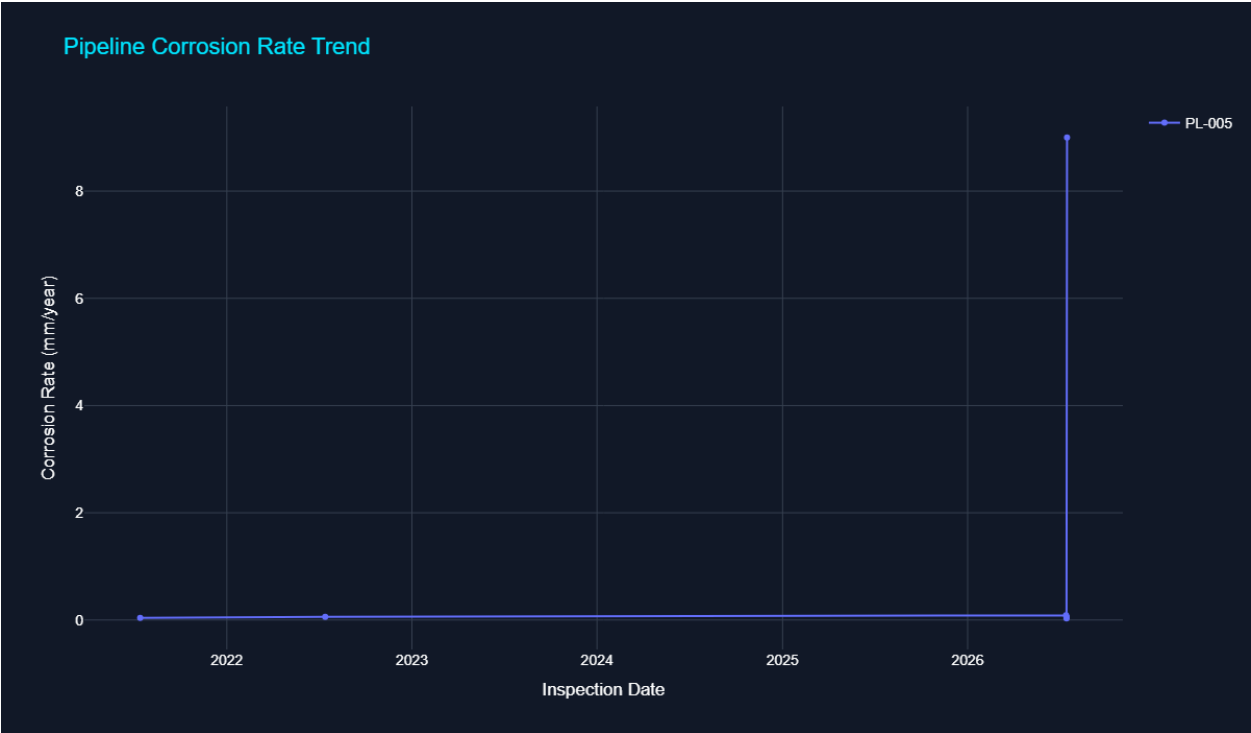
Parameter	Value
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Corrosion Rate	9.00 mm/year
Minimum Thickness	1.61 mm
Remaining Life	3.15 years
Safety Factor	18.69
Corrosion Loss	60.00 %
Health Score	38.00 %
Failure Year	2029

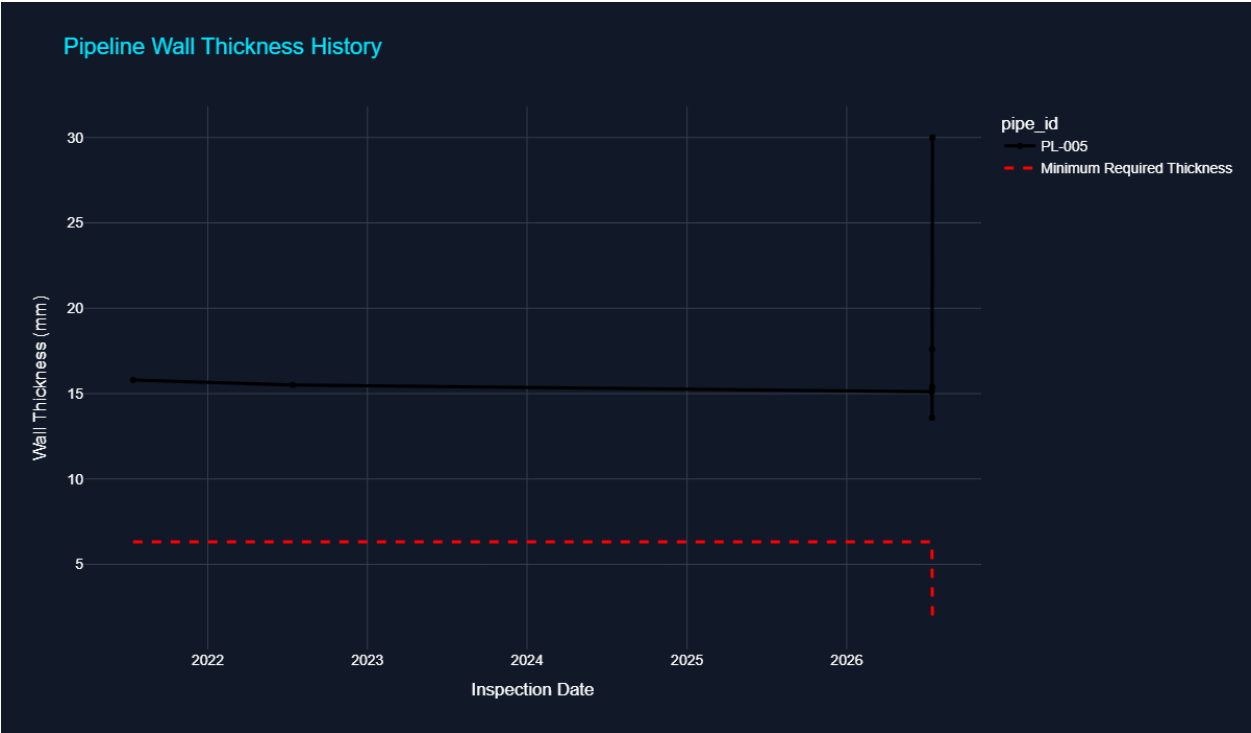
KPI Summary

Health Score	38.00%
Risk Level	Critical
Remaining Life	3.15 Years
Corrosion Rate	9.00 mm/year

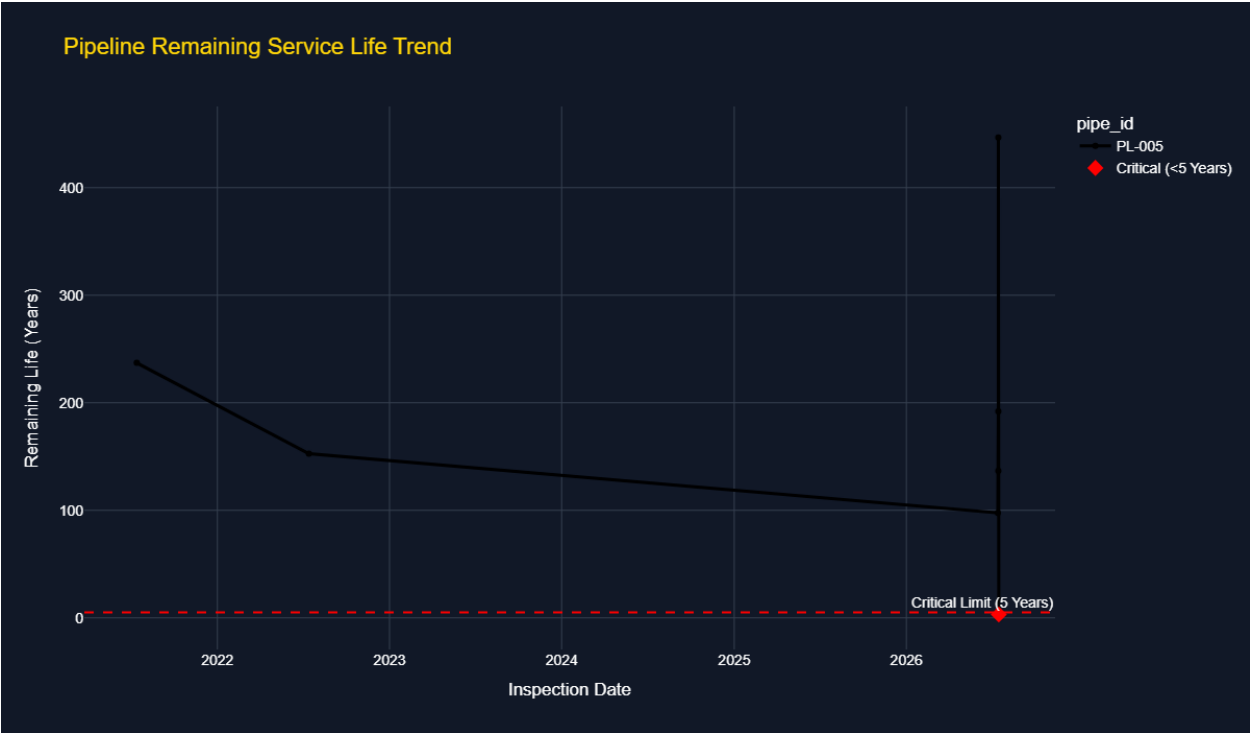
Corrosion Trend



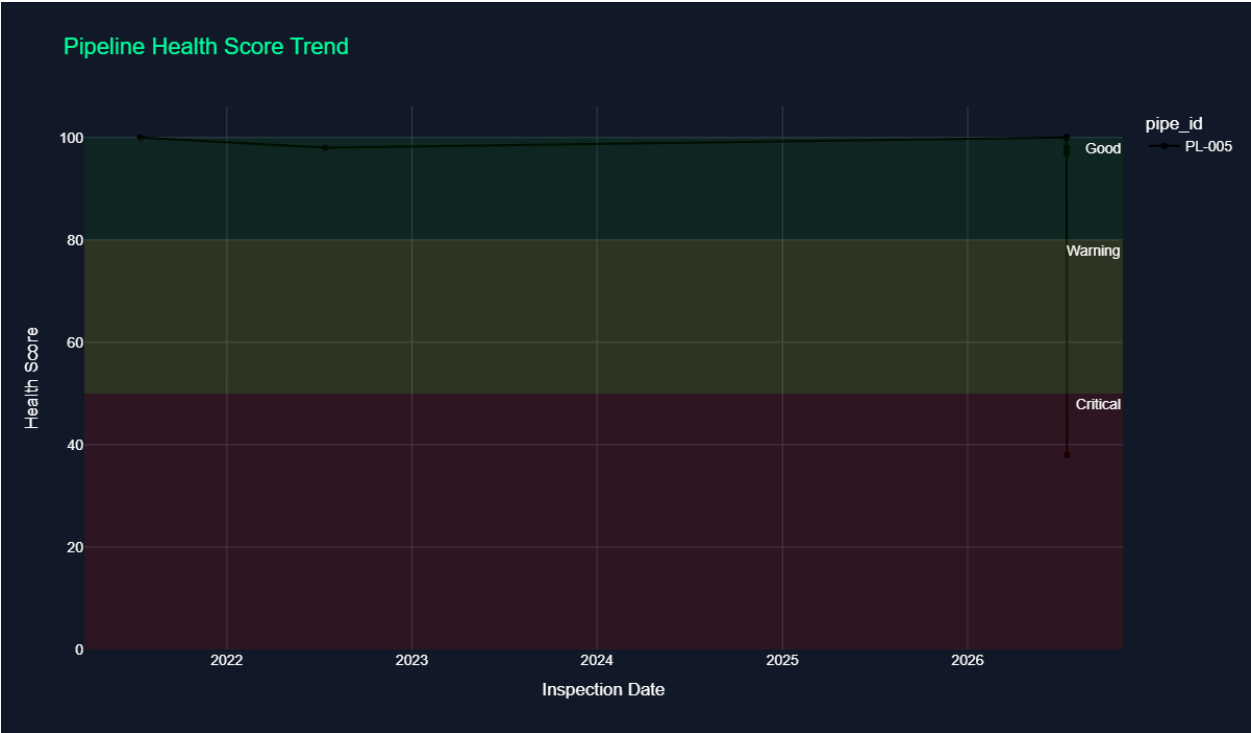
Thickness History



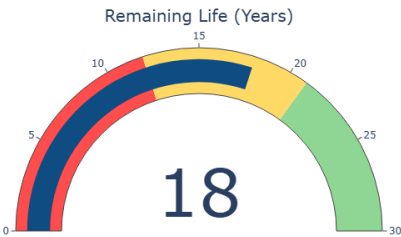
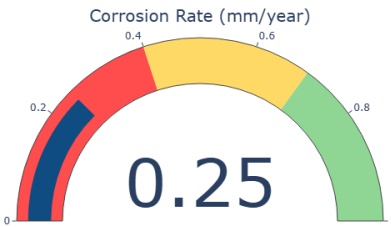
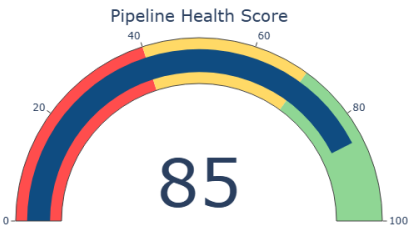
Remaining Life Trend



Health Trend



Engineering KPI Gauges

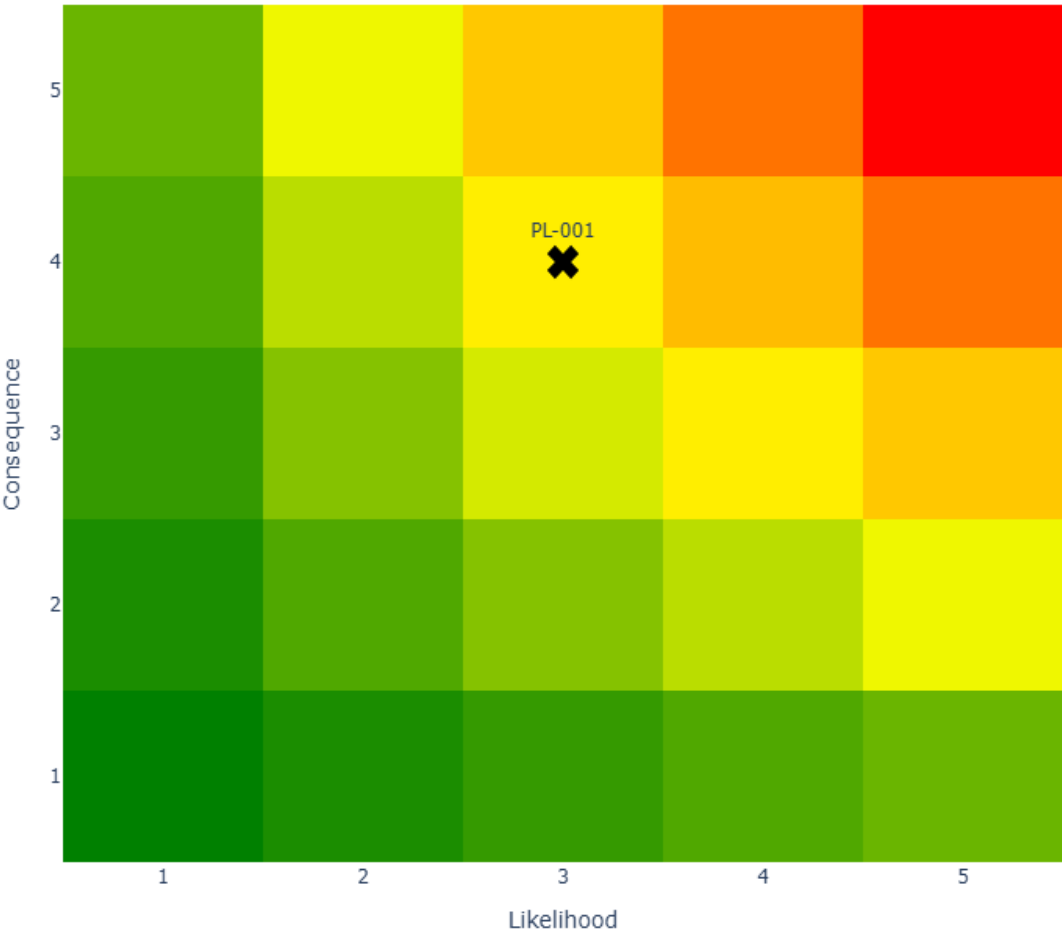


Risk Assessment

Parameter	Value
Likelihood	5
Consequence	4
Risk Score	20
Risk Category	Critical

5 × 5 Risk Matrix

5 × 5 Pipeline Risk Matrix



Engineering Recommendation

Overall Decision	Pipeline Unsafe for Service
Maintenance Suggestion	Immediate shutdown and replacement.
Inspection Interval	Immediate inspection required.
Priority	URGENT
Warning Message	Critical structural integrity risk.

Inspection Summary

The pipeline (PL-005) remains suitable for continued operation. The measured corrosion rate is 9.00 mm/year with an estimated remaining service life of 3.15 years. The calculated health score is 38.0% and the overall risk category is Critical. Immediate shutdown and replacement.

Conclusion:

Based on the inspection data and engineering calculations performed in accordance with ASME B31.3 and API 570 principles, the pipeline integrity has been evaluated and appropriate maintenance recommendations have been provided to support safe and reliable operation.

Engineering Notes

Engineer observations, maintenance remarks, or site comments may be recorded below.

Engineering Standards

- ASME B31.3 – Process Piping

- API 570 – Piping Inspection Code

The engineering calculations, integrity assessment, corrosion evaluation, remaining life estimation, and inspection recommendations presented in this report are based on the principles and methodologies defined in ASME B31.3 and API 570. The report is intended as an engineering decision-support document and should be reviewed by qualified personnel before maintenance or operational decisions are made.